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Research Article

Modelling and analysis of an 80-MW parabolic trough concentrated solar power plant in Sudan

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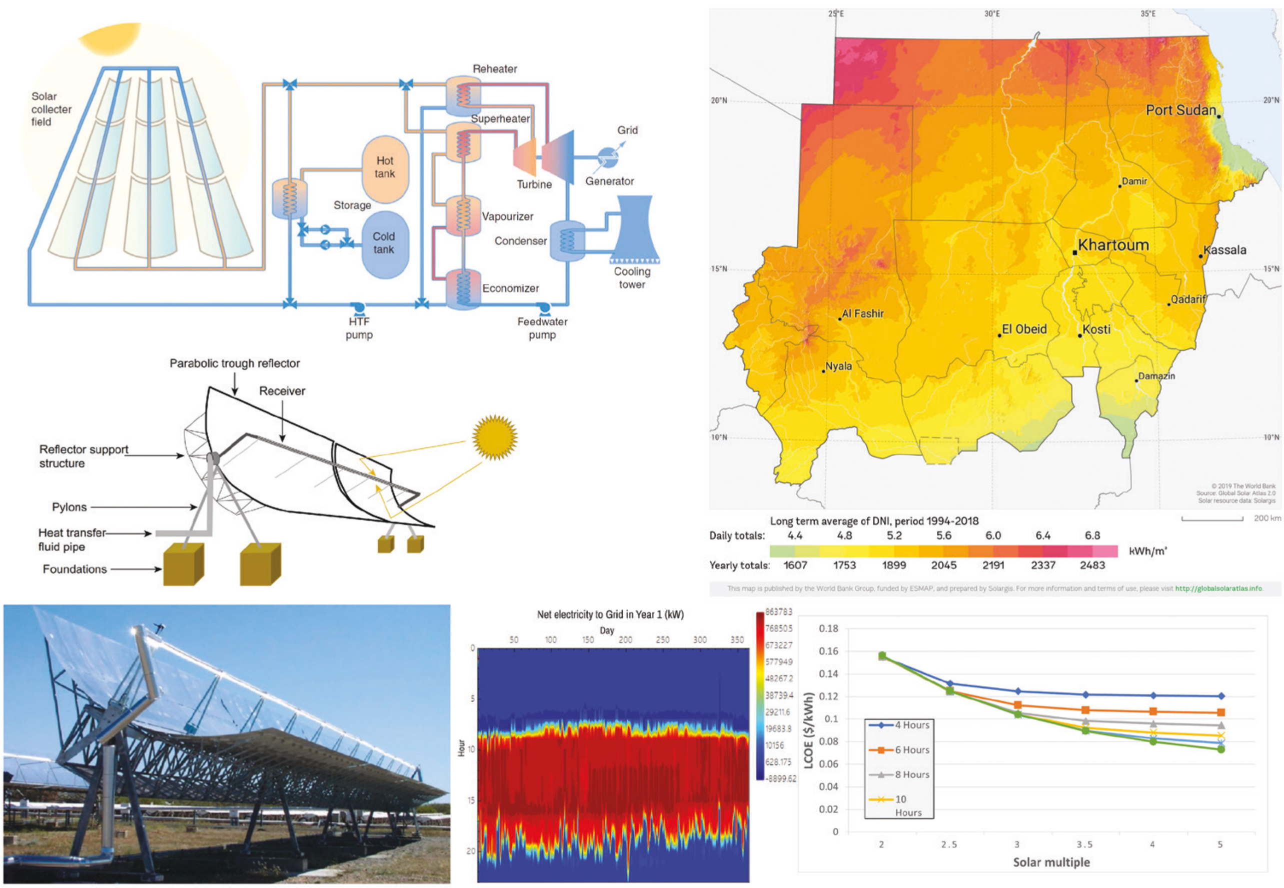
Abstract

Concentrated solar power plants can play a significant role in alleviating Sudan's energy crisis. These plants can be established and implemented in Sudan, as their potential is considerably high due to the climate conditions in Sudan. This study investigates the design of a parabolic trough concentrated solar power plant in Sudan and analyzes its technical and economic feasibility. The simulation of the plant's model used System Advisor Model (SAM) software. To determine the best location for the construction of the plant, data from 15 cities in Sudan were compared with each other based on their solar radiation and land properties. Wadi Halfa, a city in the northern region of Sudan, was chosen as the location due to its good topographical properties and climate conditions. The results show that the proposed plant can generate 281.145 GWh of electricity annually with a capacity factor of 40.1% and an overall efficiency of 15%. Additionally, a simple cost analysis of the plant indicates a levelized cost of electricity of 0.155 $/kWh. As the study results are consistent with the characteristics of similar plants, the proposed plant is considered technically and economically feasible under the conditions at its location.

Graphical Abstract



Keywords: solar energy; concentrated solar power; parabolic trough collector; System Advisor Model (SAM) simulation

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Introduction

Renewable-energy technologies are gaining more importance every day, and they are getting closer to being the leading and most influential factor in the energy production sector. The reasons for this are the negative effects of conventional energy production methods. Because of the continuous growth of the world's population and the rapidly growing industrial process, energy production rates are increasing in an unprecedented way to cover these needs. The International Energy Agency (IEA) statistics show that electricity consumption worldwide increased from 10 896.9 to 25 027.3 TWh between 1990 and 2019, which means that it increased at a rate of 129.7% [1]. Most of the electricity generated worldwide (≅62%) in 2019 was produced from primary fuels such as oil, coal and natural gas. In addition, nuclear power plants were responsible for producing ~10% of the electricity in the same year [1]. The massive use of conventional energy sources has led to dangerous levels of CO_2 emissions. In 2019, 14 068 Mt of CO_2 emissions from electricity and heat production processes were recorded globally, at an increment rate of 84.6% from the level recorded in 1990 [1]. Therefore, it is evident that problems related to the use of these conventional energy sources, such as global warming, depletion of the ozone layer and air pollution, are becoming increasingly serious, making the trend towards renewable energy very urgent and vital. However, renewable energy sources are not limited in quantity, unlike conventional energy sources, which are threatened by rising consumption. All these reasons make renewable-energy technologies the preferred choice for developing countries facing energy crises and for developed countries trying to provide solutions to the environmental risks caused by conventional energy sources.

One of the most important renewable energy sources, solar energy, can be implemented using photovoltaic (PV) panels and concentrated solar power (CSP) technologies. These two methods have different working principles. While PV systems use the principle of the photovoltaic effect to convert the sun's light into electrical energy directly, CSP technologies take advantage of the sun's heat to generate electricity by using thermodynamic cycles and heat engines. The utilization of PV systems in energy production is much larger than that of CSP technologies. In 2020, the total installed capacity of operating PV solar power plants around the world was 709 674 MW and the share of these plants in renewable energy generation by renewables was 24.3%. During the same year, the total installed capacity of CSP plants was 6479 MW and their share in renewable energy generation was 0.2% [2].

As shown in Fig. 1, CSP plants are classified into four groups depending on the type of solar concentrator used to collect and concentrate solar radiation [3]. Parabolic trough collectors (PTC) and linear Fresnel reflector solar collectors concentrate the solar radiation they receive on the absorber tube, which runs along with the collector configuration. Therefore, they are called linear concentrators. On the other hand, parabolic dish and central receiver collectors are called point-focus concentrators because they redirect solar radiation to a specific point where the receiver is located [4]. Most existing CSP plants are either parabolic trough or central receiver types. Parabolic trough concentrated solar power (PTCSP) plants are the most common, with 82 in operation worldwide, followed by central receiver CSP plants, which are in operation at 31 locations [5].

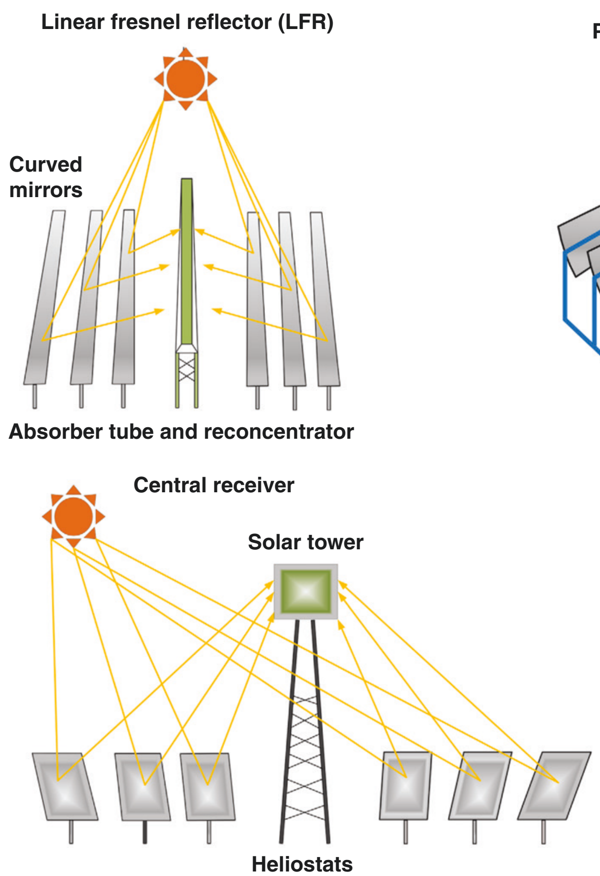

Fig. 1: Solar concentrators (used with permission from John Wiley and Sons) [3].

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Sudan, one of the developing countries, faces a massive energy crisis. Only 54% of Sudan's population had access to electricity in 2019 [6]. Most of the electricity in Sudan is generated using oil-fired thermal power plants and hydroelectric plants, with a small share from solar PV systems and solid biofuels [1, 7]. In 2020, the total installed capacity of PV systems in Sudan was 17.87 MW, which—for comparison—represents only 0.97% of the installed capacity of hydroelectric plants recorded in the same year [8]. Furthermore, the electricity production sector in Sudan does not yet include the implementation of CSP plants, although there is a very high potential for such systems to be used [8].

Taking into account the facts explained above, it is clear that CSP plants can play a significant role in providing solutions to the Sudan energy crisis. This work explores the possibility of using CSP plants to cover energy demands in Sudan by studying the design of a hypothetical 80-MW PTCSP plant and modelling it using System Advisor Model (SAM) software.

1 Literature review

Most of the research on solar energy in Sudan has focused on PV systems. Abdalla and Özcan discussed a possible design of a 1-GW_p grid-connected solar PV plant in Dongola, Sudan. The modelling and simulation of the proposed plant were performed using PVsyst software. It was found that this plant could meet the electricity needs of 7.4 million people and reduce carbon emissions by 18 million tons of CO_2 per year [9]. Similar work by Abdeen *et al.* was also carried out to study the productive energy of a 5-MW grid-connected solar PV plant in 10 different areas in Sudan. A PV plant in the proposed location of Port Sudan, a city in eastern Sudan, was found to produce 8 566 750 kWh of electricity annually [10]. In the same way, Abuelyamen and Mohamed investigated the installation of a 10-MW solar PV plant in Dongola, Sudan. RETScreen v.4 software was used to simulate the feasibility and economic analysis of the proposed plant. The study demonstrated that the proposed plant could produce ≤21.828 GWh of electrical energy and reduce ≤21 600 tons of CO_2 emissions annually [11]. Bakhit suggested decentralized power generation using PV and dish-Stirling systems as a solution to the energy crisis in Sudan. Using SAM software, both systems were analysed and compared to determine the most efficient and economically feasible. The study concluded that the PV system was more cost-competitive than the dish-Stirling engine in the climate conditions of Sudan [12].

Research related to the topic is very rare. Only a few studies explored the methods of designing PTCSP plants in Sudan. In a study conducted by Azouzoute *et al.*, meteorological data from 13 countries in the Middle East and North Africa (MENA) region were used to analyse the performance of a hypothetical 50-MW_e PTCSP power plant using Greenius simulation software. The study results showed that the proposed plant in Sudan could produce 231.5 GWh annually with a levelized cost of electricity (LCOE) of ~11.75 c€/kWh, which is one of the lowest in the MENA region [13]. Mohamed and Yahiya studied the possible options for hybridizing the Garri combined cycle power plant in Algaili, Sudan, by adding a 50-MW PTC solar field. The thermodynamic analysis performed in the study showed that the proposed solution could increase the plant's capacity by ≤90 MW [14]. In a study conducted by Ibrahim and Khayal, a 100-MW PTCSP plant was modelled using SAM software for six different cities in the River Nile state in north-eastern Sudan. The model analysis showed that the best performance could be achieved in Abu Hamad City with a capacity factor of 0.45, an annual energy generation of 389.45 GWh and a reduction of 76 000 000 m^3 of CO_2 emissions and 31 000 000 m^3 of natural gas savings [15].

A body of research has extensively focused on CSP plants for other countries. Asiri and Suliman introduced an economical and energy analysis using SAM software for a 100-MW PTCSP plant in the Qassim region in the central part of Saudi Arabia [16]. Direct normal irradiation (DNI) at the chosen location was high (≤7.74 kWh/m^2/day in July). The solar field used 16 290 solar collectors (Siemens Sunfield 6) with a total reflective area of 893 800 m^2. The solar receiver (Siemens UVAC 2010) was selected as the absorber tube and the heat transfer fluid (HTF) was synthetic oil (Therminol VP-1, working temperature 293–391°C). A two-tank storage system with a total thermal capacity of 1870.79 MWh_t using molten salt (HITEC solar salt, working temperature 238–593°C) as a storage medium was used to store thermal energy for 6 hours per day. The power block was designed to generate energy using a steam Rankine cycle (working temperature 293–391°C) with an air-cooled condenser. The simulation showed that the plant could generate 324 781 MWh of electrical energy in the first year and ~7 927 623 MWh over the 25-year project lifetime. Furthermore, the capacity utilization factor of the plant was found to be 37.1%. The economic analysis showed the net capital cost of the proposed plant as US$485 469 888, the simple payback period as 20.5 years and an LCOE of 12.21 ¢/kWh. As a result, the study concluded that the proposed plant was economically feasible at the proposed site [16].

Similarly, Benhadji Serradj *et al.* studied the design and performance analysis of a 100-MW PTCSP plant under climate conditions at Tamanrasset, Algeria. The DNI in this location was very high, ≤2800 kWh/m^2 annually, enabling the plant to cover 78% and 60% of the city's electricity requirements during the winter and summer seasons, respectively. The solar field of the plant consisted of 1168 solar collectors (EuroTrough ET 150) and 42 048 receivers (SCHOTT PTR 70) with a total reflective area of 954 840 m^2. The solar field used synthetic oil (Therminol VP-1). A two-tank storage system (thermal capacity 1665 MWh_t) using molten salt (HITEC solar salt) was used to supply the plant with thermal energy for 6 hours per day. Two steam Rankine cycle-based power block configurations were examined: air-cooled and evaporative condensers. The thermal analysis results showed that the evaporative condenser increased the capacity factor (≤48.3%) and the solar efficiency (≤15.3%). Cost analysis showed that the plant was economically feasible with an LCOE of ~0.062 US$/kWh, a payback period of 8.78 years and a benefit/cost ratio of 1.73 [17]. The performance of a hypothetical 100-MW PTCSP plant under the climate conditions of Udaipur, India, was studied using SAM software by Bishoyi and Sudhakar. The plant's proposed location received an annual DNI of ~2248.17 kWh/m^2. The solar field of the plant consisted of 194 loops with 1552 PTC collectors. Due to its excellent thermodynamic properties, HITEC solar salt was selected as an HTF in the solar field and as a storage medium in the storage system. A power block using an organic Rankine cycle was used to produce a net energy output of 285.28 GWh annually. The plant's capacity utilization factor was calculated to be 32% and its overall efficiency was found to be ~21%. The study concluded that the proposed plant was technically feasible under the given conditions [18].

Following the same analysis method, Liaqat *et al.* discussed the design and thermal performance of a 100-MW PTCSP plant with a 6-hour heat-storage system in Nawabshah, Pakistan, using SAM software. This location was considered the optimum site

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for the plant because it received a high amount of DNI of ~1955 kWh/m^2 annually. HITEC solar salt was used in both the solar field and storage tanks. The power block of the proposed plant was designed to generate energy through a Rankine steam cycle with an evaporative condenser. The results showed that the plant could generate 245.68 GWh of electricity annually with a capacity factor of 28.1%. Furthermore, the analysis concluded that the study could be used to further investigate similar technologies in Pakistan's climate [19].

Different simulation tools were used to simulate PTCSP plants in several studies. Siqueira *et al.* used TRNSYS software to study the performance of a hypothetical 30-MW PTCSP plant in Brazil. The analysis confirmed that the best place to establish such systems was in the cities of the northern region of Brazil [20]. Kordmahaleh *et al.* used MATLAB® software to predict the output of a 25-MW proposed PTCSP plant in Yazd, Iran. It was assumed that the power was generated by a 27.8% efficient steam Rankine cycle. The analysis showed that the use of a two-tank molten salt storage system would increase the total operating time of the plant from 1726 to 3169 hours per year and increase the maximum thermal useful energy that could be obtained from the plant up to 7974 GJ annually [21].

With the increasing adoption of CSP technology, there is a significant need to explore the possibility of implementing these technologies in Sudan. Despite the wealth of literature available in the field, there is a lack of research focused on Sudan's climate and energy requirements. Although some efforts have been made to address this issue, more research is still needed. A closer look at the literature reveals several gaps and the need to conduct a comprehensive study that considers the technical details and economic benefits of PTCSP plants in Sudan. Previous studies either chose a specific location in Sudan to study the potential of PTCSP plants or generally compared the potential of PTCSP plants in some of Sudan's cities with other cities in neighbouring countries. This study covers all regions of Sudan and determines the best location for the establishment of PTCSP plants. This work is carried out in response to the need to fill the research gap in CSP technologies in the conditions of Sudan.

2 Methods and modelling

The SAM software is used to design, simulate and analyse the hypothetical PTCSP plant. The National Renewable Energy Laboratory (NREL) developed this software to help project managers, engineers and researchers model different renewable-energy systems such as CSP plants, PV systems, wind turbines and geothermal power applications. Additionally, SAM offers economic analysis for systems that facilitate the decision-making process.

SAM software uses two approaches to simulate PTCSP plants: physical and empirical models. The empirical model is based on real scenarios from several existing plants. On the other hand, the physical model has more flexibility and allows users to change the design parameters [22]. This model can be validated by comparing the simulation results with the actual measured data from PTCSP plants. For small PTCSP plants, data from a 50-KW pilot plant established for research purposes in Louisiana, USA, were used to validate the model. This plant used an organic Rankine cycle to produce ~22 301 kWh of electricity per year. The SAM physical model predicted this value as 22 913 kWh, ~3% higher than the actual value. Furthermore, it was found that the model predictions were correct ~97% of the year [23]. For large PTCSP plants, the model was validated using the Andasol-1 plant data. This plant was located in Aldeire, Spain and had a nominal capacity of 50 MW$_e$. The plant's thermal energy-storage (TES) system could store thermal energy for 7.5 hours using a molten salt mixture of 60% sodium nitrate and 40% potassium nitrate. The actual energy output of the plant was 179 103 000 kWh, which was 2.6% above the value obtained from the simulation. The capacity factor estimated by the model was 40.2%, which was less than the actual value by 1.3%. The actual total installed cost of the plant was $418 440 431, which was greater than the value estimated using the model by 1.6% [24]. The model was also validated using the actual data from the Genesis PTCSP plant located in Blythe, CA, USA. This plant could produce 580 GWh of electricity annually using two 125 power units. Validation was done using only half of the solar field and one of the two turbines of the power block. The monthly average values predicted by the model were found to always be close to the actual values [25].

Due to the simple design and widespread implementation of PTCSP plants, the results of the simulations were found to be relatively close to the measured operating data. Therefore, the SAM parabolic trough physical model is reliable in predicting the performance of PTCSP plants [25]. This study uses the physical model approach to simulate and analyse the hypothetical PTCSP plant.

2.1 System description

Generally, PTCSP plants consist of three parts: solar field, TES system and power block (Fig. 2) [26]. The solar field is made up of multiple rows of PTC. Its function is to capture solar radiation and use it to heat the HTF that flows through the absorber tube of the collectors. The HTF is circulated inside the solar field until its temperature reaches the desired level. Once this happens, the HTF is directed directly to the power block or the TES system. The TES system is responsible for supplying the power block with heat energy under conditions with little or no solar radiation. Although the TES system can improve plant performance with load control and generation flexibility (dispatchability), it is not an essential component of the plant. The two-tank molten salt TES system is the most used in PTCSP plants. This kind of system is used to keep molten salts at high temperatures during specific daytime hours. Heat exchangers transfer thermal energy between the HTF of the solar field and the storage medium or the working fluid of the power block. The power block converts thermal energy into mechanical energy and then into electricity through thermodynamic cycles. Steam and organic Rankine cycles are the most used in this type of power plant.

2.2 Site selection and solar resource

When selecting the location of CSP plants, several aspects should be considered. These considerations are the intensity of solar radiation in the form of DNI, wind speed, availability of water resources, the land area available for construction, land cover, slope and accessibility of the electricity grid [27]. It is strongly recommended that the annual cumulative DNI value at the site be >2000 kWh/m^2. However, the site should not receive <1600 kWh/m^2 of DNI annually [4]. Strong wind speeds can lead to larger thermal losses in the solar field and impair the support of the plant components. Therefore, it is preferable to establish CSP plants where the wind speed is as low as possible [27]. A reliable water resource is critical to the operation of CSP plants, as water

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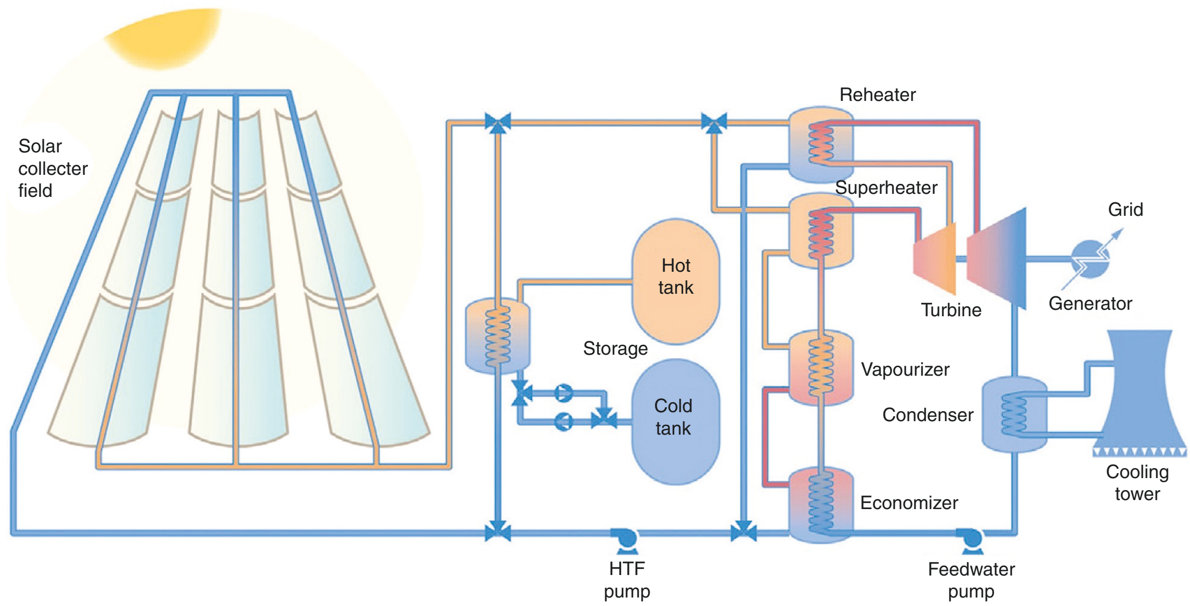

Fig. 2: Configuration of a typical PTCSP plant working with thermal oils [26].

is used to wash mirrors, replace leakage in the steam cycle and supply cooling towers [27]. The slope of the land should not exceed 3% and the surrounding area should not contain terrain that casts shadows on the solar field [4]. The plant should have access to the electricity grid to deliver the generated electricity to customers. The distance to the grid should be minimal because each kilometre of the transmission line costs more money [27].

Sudan lies between 22° to 10° N and 38° and 22° E and occupies an area of 1.886 million km^2 in the north-eastern region of Africa. Except for a small area in the east, most of the areas of Sudan receive >1700 kWh/m^2 of DNI annually. Fig. 3 shows the distribution of DNI in Sudan [28].

Data from 15 different cities in Sudan were obtained from Meteonorm 7.3 software. These data include solar radiation values, wind speeds, precipitation ratios, sunshine durations and daily temperatures. The location with the highest annual average daily DNI (Fig. 4) is Al Fashir (6.65 kWh/m^2/day), followed by Wadi Halfa (6.6 kWh/m^2/day). On the other hand, Port Sudan receives the lowest DNI value among the proposed locations.

For further analysis, Google Earth software is used to explore the topography and water resources at the locations. The characteristics of the selected locations (Table 1) were analysed to determine optimal locations to establish the plant. Although Al Fashir receives the highest DNI value, a permanent water resource (Nile River) gives Wadi Halfa an advantage. Also, among the cities with the lowest average annual wind speeds (Port Sudan, Nyala and Wadi Halfa), Wadi Halfa has the highest DNI value. Furthermore, the topography of Wadi Halfa, which is a desert, allows the plant to be constructed without affecting the natural system or being affected by the surrounding terrain. Therefore, it is logical to consider Wadi Halfa as the most convenient site to establish such a plant. The monthly global horizontal irradiation (GHI) values (Fig. 5) for Wadi Halfa show that the proposed location receives >200 kWh/m^2 of solar radiation for the 6 months from March to August.

The highest and lowest GHI values in Wadi Halfa occur in May and December, respectively, with yearly maximum and minimum temperatures of 46.6°C and 5°C (dry-bulb) and 24°C and 1.7°C (wet-bulb).

2.3 Solar field configuration

The PTC consist of two main elements: the reflector and the receiver (Fig. 6) [29]. The reflector is a U-shaped mirror that captures solar radiation and reflects it onto the receiver. To increase its reflective properties, aluminium or silver-coated glass with a protective copper layer is used to manufacture most types of PTC [30]. The receiver is a tube that runs across the multiple collectors to absorb the heat of the solar radiation reflected by the reflectors. Receivers are usually made of stainless steel, covered with a black coating and surrounded by an evacuated glass tube. With this manufacturing process, the receiver achieves high absorptive efficiency and the heat losses are reduced to the minimum limit [31].

The essential geometric characteristics of the reflector (Fig. 7) are the focal length, aperture width and length. Focal length is the distance between the vertex of the reflector and the focal point. Aperture width (w_a) refers to the distance between the collector rims. The aperture area, which determines the amount of radiation that the collector can capture, is given by Equation 1 [32]:

$$A_p = w_a \cdot L \tag{1}$$

where L represents the collector length (m). The concentration ratio is a measure of the reflector size compared to the receiver size, which in the case of PTC, can be given by Equation 2 [32]:

$$C = \frac{A_p}{A_a} = \frac{w_a}{\pi D_a} \tag{2}$$

where A_a and D_a represent the absorber's area and the outside diameter, respectively.

The optical efficiency (η_o) of a collector is defined as the ratio of the energy absorbed by the receiver to the solar radiation incident on the reflector aperture [32]. The optical efficiency of the collector decreases due to manufacturing errors, wrong construction, imperfection of the collector materials and external factors such as dust and rain [31]. The total energy absorbed by the receiver tube (Q_t) can be expressed by Equation 3 [32]:

$$Q_t = A_p \eta_o G_b \tag{3}$$

where G_b represents the solar beam irradiance incident on the collector (W/m^2), which is calculated from Equation 4 [4]:

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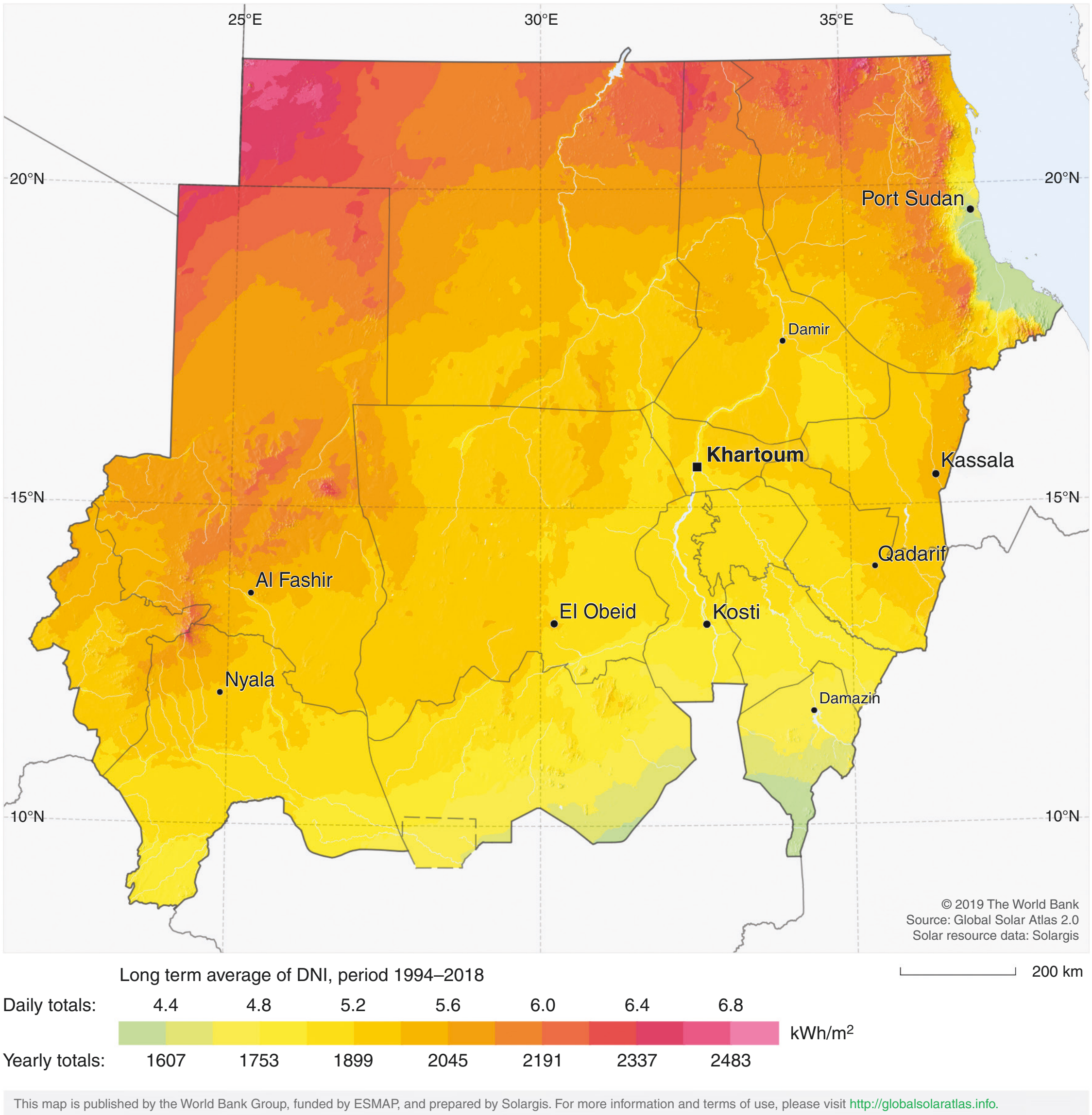

Fig. 3: DNI distribution in Sudan [28].

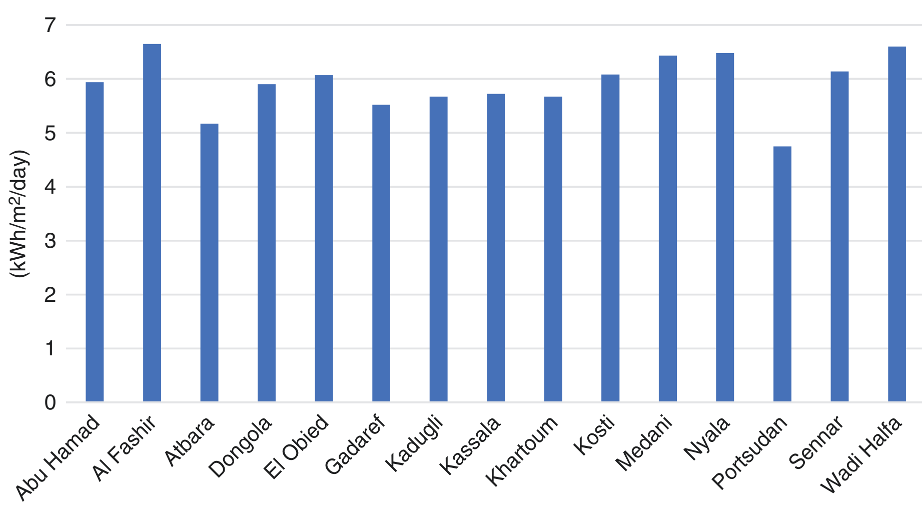

Fig. 4: Annual average daily DNI in 15 cities in Sudan (Source: Meteonorm 7.3).

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Table 1: Characteristics of 15 proposed cities

City	Coordinates	Annual cumulative DNI (kWh/m²)	Average annual wind speed (m/s)	Available water resource	Land cover
Abu Hamad	19.53° N, 33.3° E	2168.1	4.9	Perennial[a] (Nile River)	Desert
Al Fashir	13.63° N, 25.36° E	2427.25	4.2	Groundwater[c]	Desert
Atbara	17.7° N, 34° E	1887.07	4.3	Perennial[a] (Nile River) seasonal[b] (Atbara River)	Desert
Dongola	19.17° N, 30.5° E	2153.5	4.4	Permanent (Nile River)	Desert
El Obied	13.18° N, 30.17° E	2215.55	4.3	Groundwater[c]	Desert
Gadaref	14.02° N, 35.4° E	2014.8	4.3	Groundwater[c]	Agricultural land
Kadugli	11° N, 29.7° E	2069.55	4.3	Groundwater[c]	Forests
Kassala	15.4° N, 36.42° E	2087.8	4.3	Groundwater[c], seasonal[b] (Alqash River)	Agricultural land
Khartoum	15.5° N, 32.56° E	2069.55	4.3	Perennial[a] (Nile River)	Desert
Kosti	13.14° N, 32.67° E	2219.2	4.3	Perennial[a] (White Nile River)	Agricultural land
Medani	14.4° N, 33.53° E	2346.95	4.3	Perennial[a] (Blue Nile River)	Agricultural land
Nyala	12.02° N, 24.83° E	2365.2	3.2	Groundwater[c]	Desert
Port Sudan	19.6° N, 37.2° E	1733.75	3.2	Perennial[a] (Red Sea and desalination plants)	Mountains
Sennar	13.5° N, 33.6° E	2241.1	4.3	Perennial[a] (Blue Nile)	Agricultural land
Wadi Halfa	21.8° N, 31.35° E	2409	3.9	Perennial[a] (Nile River)	Desert

[a]Permanent surface water.
[b]Water that is present in the spring and disappears by summer.
[c]Underground water surfaces.

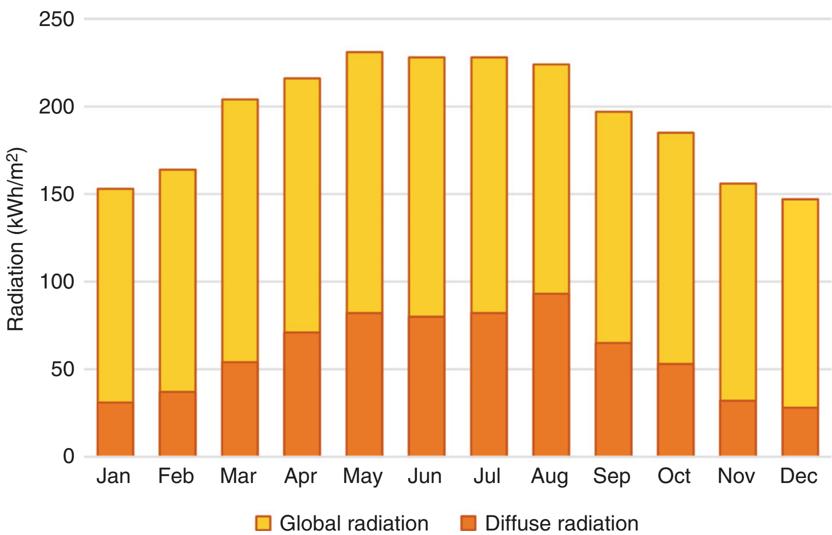

Fig 5: Monthly GHI at Wadi Halfa (Source: Meteonorm 7.3).

$$G_b = DNI \cdot \cos(\theta) \tag{4}$$

where θ represents the incidence angle of solar radiation. *DNI* in Equation 4 should be in terms of W/m².

Another factor that may cause efficiency reduction is heat loss from the absorber to the ambient air. This heat loss occurs due to convection, radiation and conduction heat transfer processes (Fig. 7). The convection loss from the tube to the glass cover can be neglected because of the vacuum gap between the tube and the cover. In addition, the conduction loss through the support brackets is small compared to other losses and its effect can be neglected, too. The heat loss coefficient (U_L) is given by Equation 5 [32]:

$$U_L = \left(\frac{A_r}{(h_w + h_{r,c-a})\, A_c} + \frac{1}{h_{r,r-c}} \right)^{-1} \tag{5}$$

where h_w represents the convection loss to the ambient air (W/m²·K), $h_{r,c-a}$ represents the radiation loss from the glass cover to the ambient air (W/m²·K), $h_{r,r-c}$ represents the radiation loss from the tube to the glass cover (W/m²·K) and A_r and A_c represent the cross-sectional area of the tube and cover (m²), respectively.

The useful energy (Q_u) is the energy transferred to the HTF that is circulated inside the receiver tube. It can be calculated using Equation 6 [32]:

$$Q_u = A_p \eta_o I_b \cos(\theta) - A_r U_L (T_r - T_a) \tag{6}$$

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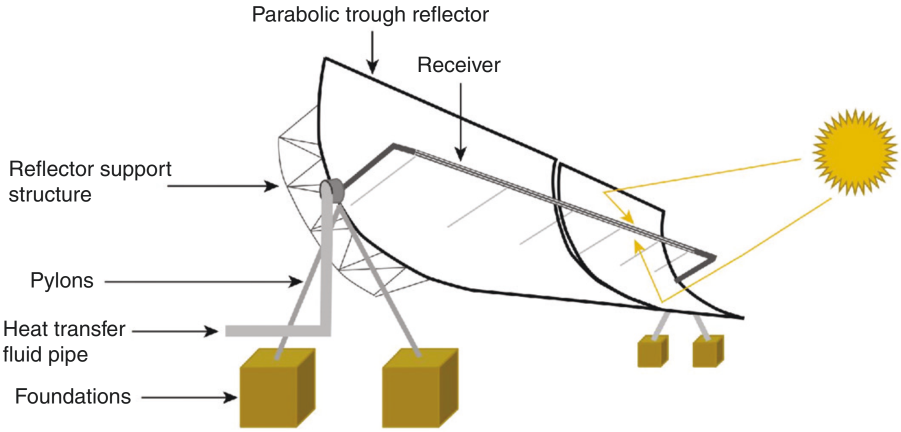

Fig. 6: PTC configuration (used with permission from Elsevier) [29].

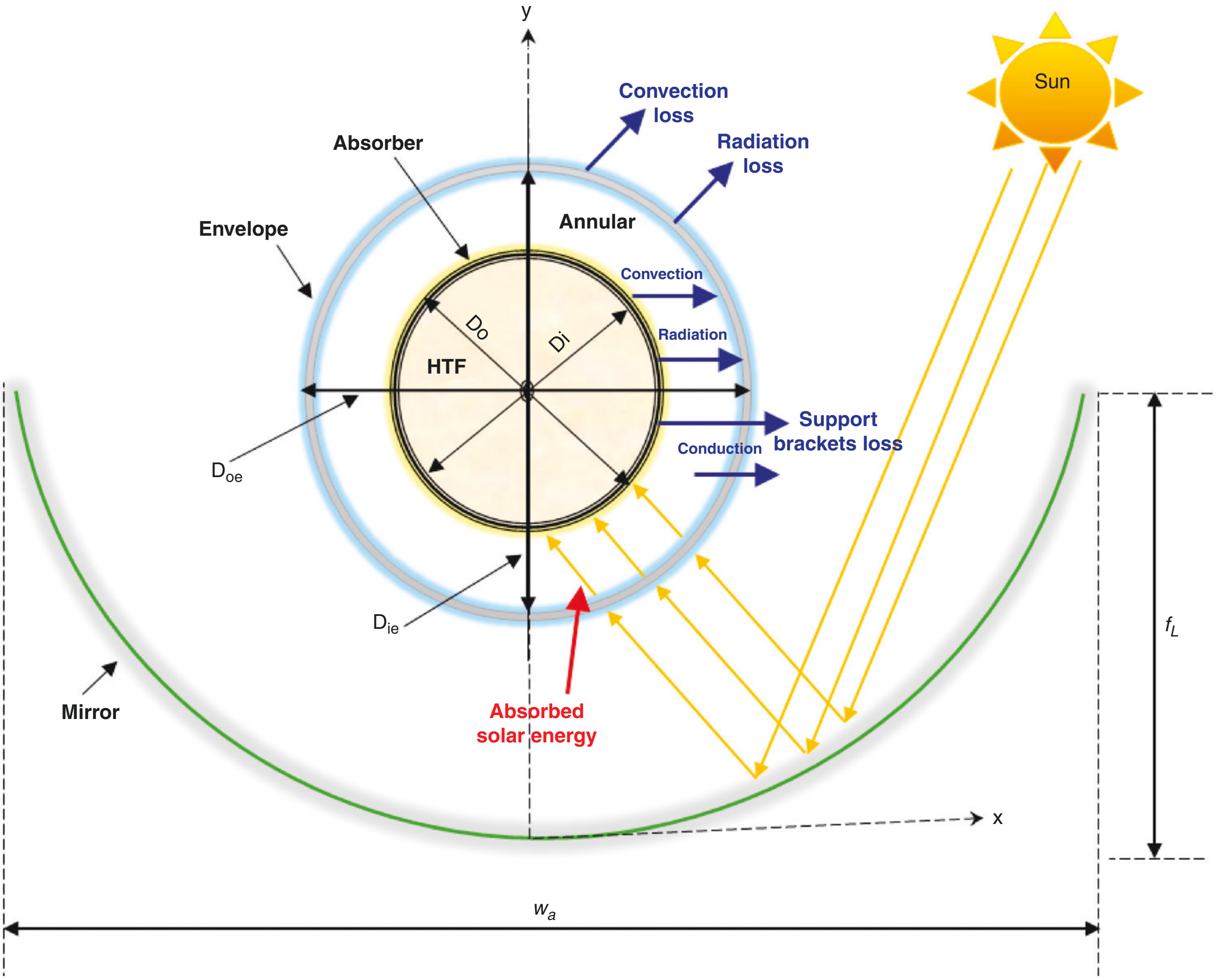

Fig. 7: PTC geometry and working principle [33].

where T_r and T_a represent the temperatures of the receiver tube and ambient air, respectively.

The overall efficiency of the collector (η_c) is defined as the ratio of the useful energy to the total absorbed energy. It can be expressed as in Equation 7 [32]:

$$(\eta_c) = \frac{Q_u}{Q_t} \tag{7}$$

In this study, EuroTrough ET150 (Fig. 8) is selected to form the collector of the solar field. EuroTrough PTC types have several advantages over the other PTC types, such as durability and lightness of the structures, fewer required components, simple manufacturing and lower cost. For all of these reasons, this type of PTC is considered very competitive and efficient to be used in PTCSP plants [30].

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SCHOTT PTR70 (Fig. 9) is chosen as the solar field receiver. This type of receiver is characterized by its stable performance and higher efficiency by virtue of low heat loss.

The solar field in the PTCSP plants (Fig. 10) is divided into several major subsections. Each subsection consists of many loops that connect a number of collectors with a single receiver tube. Hot header pipes collect the HTF from each subsection and send it to the power block or to the TES system. Cold header pipes do the reverse process.

Fig. 8: EuroTrough ET 150 (used with permission from Elsevier) [34].

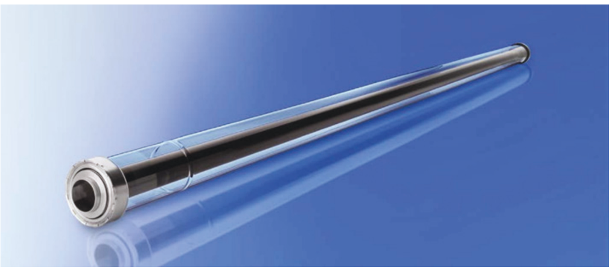

Fig. 9: SCHOTT PTR70 [35].

The solar field of the PTCSP plants can be sized by the solar multiple, which refers to the ratio of the actual aperture area of the field to the aperture area required to operate the plant at its design capacity [18]. Optimizing the solar multiple is essential for CSP system design. The optimal value of the solar multiple is within the range of 1.15–1.3 for plants without storage systems, while for plants containing storage systems, it should be >2 [37]. Since the proposed plant includes a TES system, the solar multiple is assumed to be 2.

To obtain the maximum output of the solar field, one-axis tracking systems are used. There are two possible choices of orienting solar fields: north–south orientation with an east–west tracking technique or east–south orientation with a north–south tracking technique. The north–south orientation is considered more efficient and has a better behaviour than the east–west orientation [16]. Therefore, the solar field of the proposed PTCSP plant is designed to have a north–south orientation with an east–west tracking system. Table 2 summarizes the characteristics and design assumptions of the solar field and its components.

2.4 HTF selection

Both molten salts and thermal oils can be used as the HTF in the solar field of PTCSP plants. However, because of the high freezing point of the molten salts, more power is consumed in molten salt CSP plants to maintain fluidity and prevent freezing, raising the costs and making the system more complex. Another disadvantage of using molten salts as HTFs is the high heat losses due to the high working temperatures of molten salts. For these reasons, thermal oils are used in most operating PTCSP plants [38]. Therminol VP-1 is chosen as the HTF in the solar field of the proposed plant. Table 3 lists the physical and thermal properties of the selected HTFs [39].

The HTF is assumed to work within a temperature range of 293–393°C. The HTF is heated up in the solar field until its temperature reaches 393°C and then sent to the power block. To prevent fluid freezing, the freeze protection temperature is set to be 40°C. The HTF temperature is not allowed to go below this temperature. The assumptions for maximum and minimum single loop flow rates are 1 and 12 kg/s, respectively.

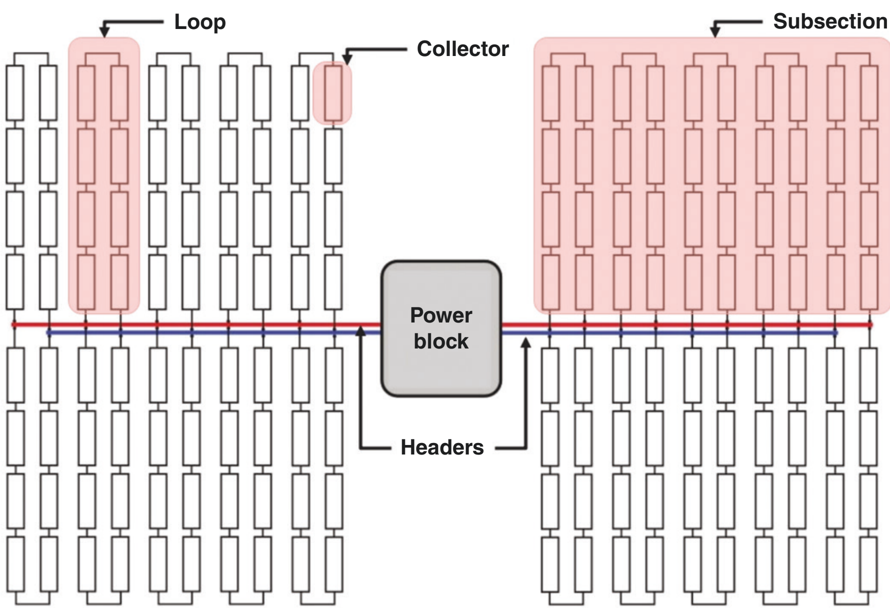

Fig. 10: An example of solar field layout in PTCSP plants [36].

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Table 2: Summary of the solar field characteristics

Reflector (EuroTrough ET150)	
Aperture area	817 m^2
Focal length	2.11 m
Optical efficiency	87.1 %
Receiver (SCHOTT PTR70)	
Absorber tube inner diameter	0.066 m
Absorber tube outer diameter	0.07 m
Glass envelope inner diameter	0.115 m
Glass envelope outer diameter	0.12 m
Optical derate	0.87
Heat loss per metre of length	70 W/m @ 250°C 110 W/m @ 300°C 165 W/m @ 350°C 250 W/m @ 400°C
Solar field layout	
Orientation	North–south
Total number of loops	119
Number of collectors for each loop	8
Row spacing	15
Number of subsections	6
Solar multiple	2
Design point DNI	890 W/m^2
Total reflective area	778 260 m^2

Table 3: Therminol VP-1 properties [39]

Appearance	Clear, water-white liquid
Composition	Biphenyl/diphenyl oxide
Total acidity	<0.2 mg KOH/g
Crystallization point	12°C
Maximum bulk temperature	400°C
Auto-ignition temperature	601°C
Density	1070 kg/m^3 @ 12°C 999 kg/m^3 @ 100°C 694 kg/m^3 @ 400°C
Heat capacity	1.523 kJ/(kg·K) @ 12°C 1.775 kJ/(kg·K) @ 100°C 2.628 kJ/(kg·K) @ 400°C
Kinematic viscosity	5.12 cSt @ 12°C 0.986 cSt @ 100°C 0.211 cSt @ 400°C

2.5 TES system

The TES system in PTCSP plants is identified by the working principle, duration of heat storage, capacity and storage medium used. A two-tank molten salt TES system is used in the proposed plant to supply the power block with thermal energy for 6 hours. This system keeps the cold and hot storage medium separated in two separate tanks. Each tank has a total volume of 21 375.6 m^3 and an available HTF volume of 19 594.3 m^3.

Due to their high heat capacity, high boiling point, thermal stability and low cost, molten salts are more suitable for PTCSP plants [40]. HITEC solar salt is selected to be the storage medium in the proposed plant's TES system. Table 4 shows the thermal and physical properties of the selected material [40].

To prevent salt crystallization inside the tanks, 260°C and 365°C are set as the lowest temperatures for the cold and hot tanks, respectively. When the salt temperature goes below these limits, an assisting heater associated with the tank works to heat the salt again.

Table 4: Properties of HITEC solar salt [40]

Composition	60% $NaNO_3$, 40% KNO_3
Fusion point	240°C
Decomposition point	565°C
Density	2090 – 0.636·T (°C) (kg/m^3)
Specific heat capacity	1443 + 0.172·T (°C) (J/kg·K)

2.6 Power block

The proposed plant is designed to generate electricity using the Rankine steam cycle with an evaporative condensing method. The design assumptions for the cycle are derived from the default values of the characteristics of the power block in the SAM software. These values were calculated for a 10-MW regenerative Rankine cycle. The operating conditions of this power cycle match the representative operating conditions of the PTCSP plants, although its scale is smaller than most of the working plants [36]. Fig. 11 illustrates the components of the basis cycle. The design parameters of the proposed plant's power block based on the basis model are listed in Table 5.

The capacity factor (CF) is defined as the ratio of the actual output of the plant to the maximum output it can generate under design conditions. Although this value ranges from 25% to 28% for plants without TES systems, the use of TES systems increases this value by ≤70% depending on the duration of the heat-storage process [37, 41]. Equation 8 expresses the CF mathematically [38]:

$$CF = \frac{\textit{Annual electricity output}}{\textit{Maximum output that could be obtained}} \tag{8}$$

The overall efficiency of the plant (η) is identified as the fraction of solar radiation incident in the solar field that is converted to electricity. Generally, PTCSP plants have an overall efficiency of 11–16% [41]. Equation 9 shows the mathematical expression of the overall efficiency:

$$\eta = \frac{\textit{Annual electricity output}}{\textit{Annal solar radiation incident on the solar field}} \tag{9}$$

2.7 Financial parameters

SAM software offers several approaches for the financial analysis of renewable-energy systems. The simplest one is the LCOE calculator method. The LCOE is a key economic factor, which is defined as the total life-cycle cost of a plant, expressed as a cost per unit of electricity produced throughout its lifetime [16]. LCOE is important for both investors and researchers, and can be used to compare different technologies. The LCOE calculator method in the SAM software uses four parameters to calculate the annual value of LCOE, as expressed in Equation 10 [42]:

$$\mathrm{LCOE} = \frac{\mathrm{FCR} \cdot \mathrm{TCC} + \mathrm{FOC}}{\mathrm{AEP}} + \mathrm{VOC} \tag{10}$$

where *TCC* represents the capital cost ($), *FOC* represents the fixed annual operating cost ($), *VOC* represents the operating cost per unit of generated electricity ($/kWh), *AEP* represents the electricity generated annually (kWh) and *FCR* stands for fixed charge rate, which is the return per amount of investment needed to cover the investment costs.

This study uses the default values of the economic parameters in the SAM software. These values are updated to fit most cases of renewable-energy systems [43]. Taking advantage of these values,

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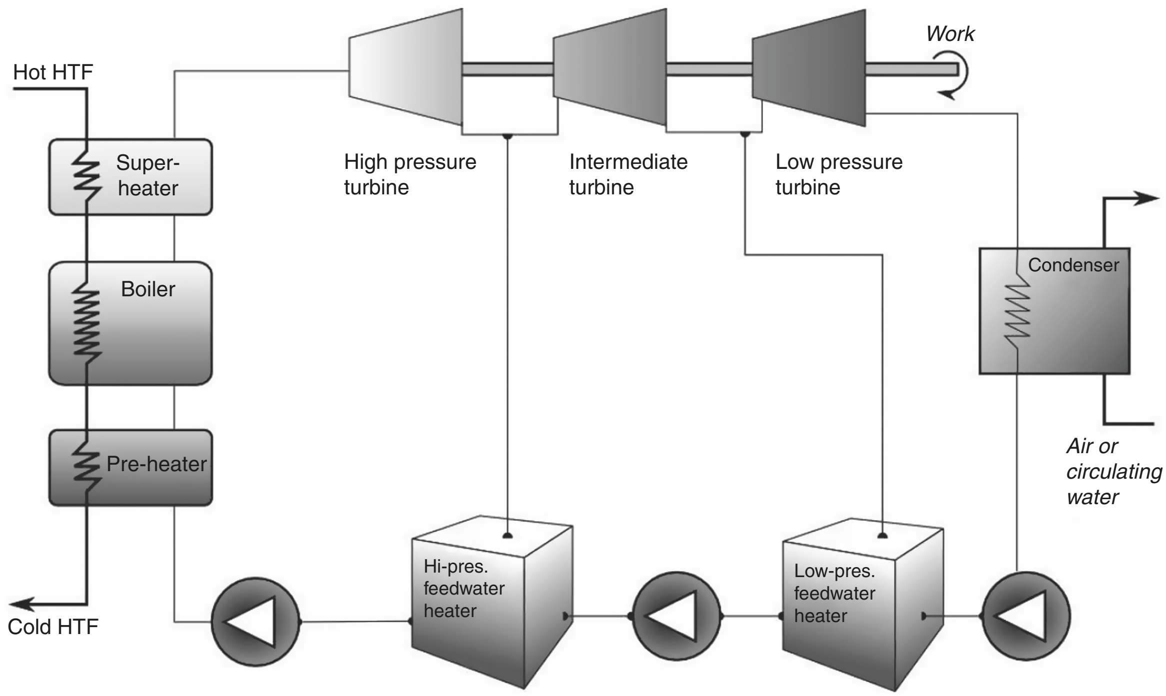

Fig. 11: Basis cycle used in SAM software [36].

Table 5: Power block design parameters

Estimated nameplate capacity	**80 MW_e**
Cycle thermal efficiency	0.356
HTF inlet temperature	393°C
HTF outlet temperature	293°C
Steam temperature at the turbine inlet	373°C
Boiler pressure	100 bar
Condenser type	Evaporative
Condenser pressure	0.085 bar
Ambient temperature at design	25°C (wet-bulb) 45°C (dry-bulb)

TCC, *FOC*, *VOC* and *FCR* are chosen to be 5627 $/kW of installed capacity, 66 $/kW of installed capacity, 0.004 $/kWh of generated electricity and 0.082, respectively.

3 Results and discussion

3.1 Thermal performance

Based on the inputs provided in the previous section, an annual analysis of the proposed plant is done. During 8760 hours of the year, the proposed plant can produce ~281.145 GWh of electricity with a CF of 40.1% and an overall efficiency of 15%. The values of the CF and overall efficiency are acceptable as they are in the range of the values of such plants. The plant uses 1.139 × 10^6 m^3 of water annually for mirror washing and evaporative condenser. This amount of water can be supplied from nearby water resources such as the Nile River. Annually, 100.033 MWh of electricity is consumed to prevent freezing, mainly in storage tanks. The key findings of the annual analysis are listed in Table 6.

Fig. 12 shows the monthly generated electricity. According to this figure, the plant generates the largest amount of electricity in March, which is ~26.41 GWh. In contrast, the lowest amount of electricity is produced in December, which is ~20.37 GWh. The variation in monthly generated electricity results from the variation in the mean incident irradiance (G_b). Due to tracking errors, this value of solar irradiance is always less than the normal beam irradiance (I_b). The difference between received and absorbed solar energy for each month (Fig. 13) shows that although the solar field receives more solar energy in January, November and December, the tracking losses are higher in these months than in March.

Table 6: The key findings of the annual analysis

Received solar thermal energy	**1874.19 GWh_t**
Net electrical energy production	281.145 GWh_e
Power cycle gross electrical output	321.012 GWh_e
Gross-to-net conversion	87.6%
Freeze protection	140 882 kWh_e
Water usage	1 139 033 m^3
Overall efficiency	15%
Capacity factor	40.1%

Generated electricity visualized in a heat map as a function of day hours (Fig. 14) shows that the plant can produce electricity for ~13 hours on most days of the year. The plant achieves its maximum output between 10 and 16 hours, which is >7 × 10^4 kW_e on most days. Therefore, the plant seems to be able to meet the electricity needs that arise during these hours, such as air-conditioning loads.

Tracking the flow of thermal energy between the different plant components for each month of the year (Fig. 15) shows that the solar field can supply the power cycle with sufficient energy to operate and charge the TES system simultaneously during the day hours. When the solar field is not in operation, the TES system works to operate the power cycle for an additional 6 hours. In March, when the plant generates the highest amount of energy, 18.304 × 10^3 MW_t of thermal energy is discharged to keep the

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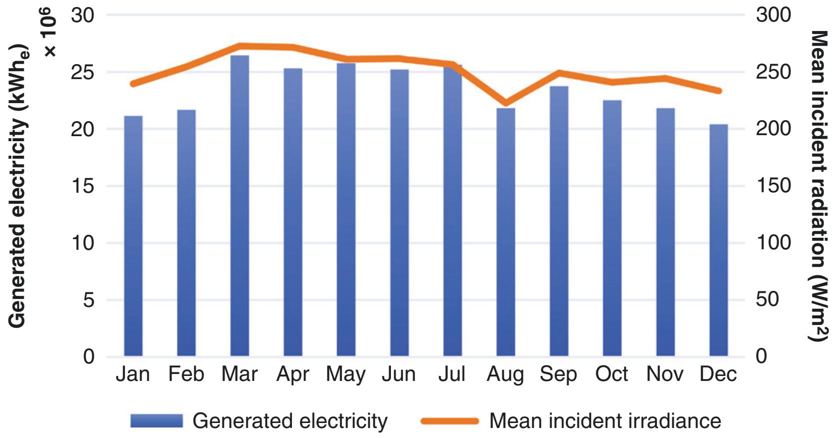

Fig. 12: Monthly generated electricity.

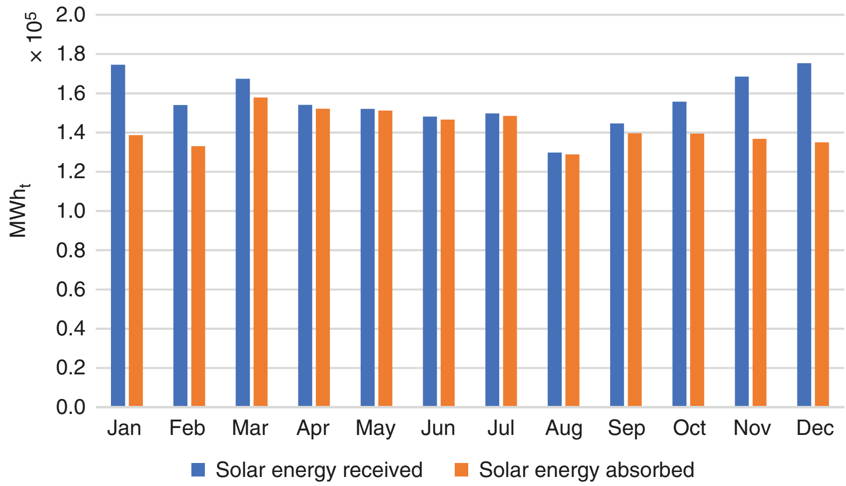

Fig. 13: Monthly solar energy received vs. solar energy absorbed.

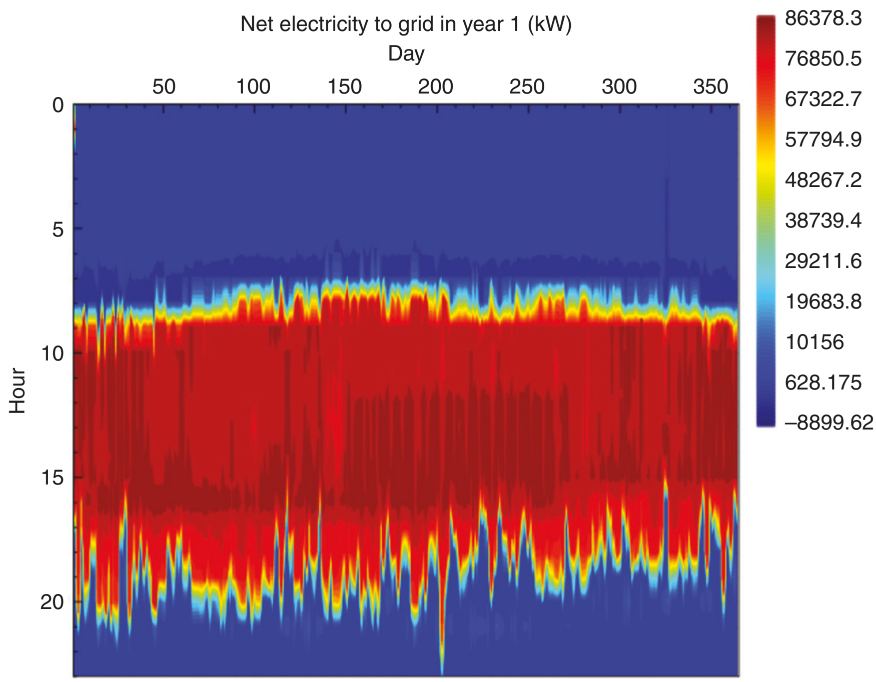

Fig. 14: Generated electricity as a function of day hours.

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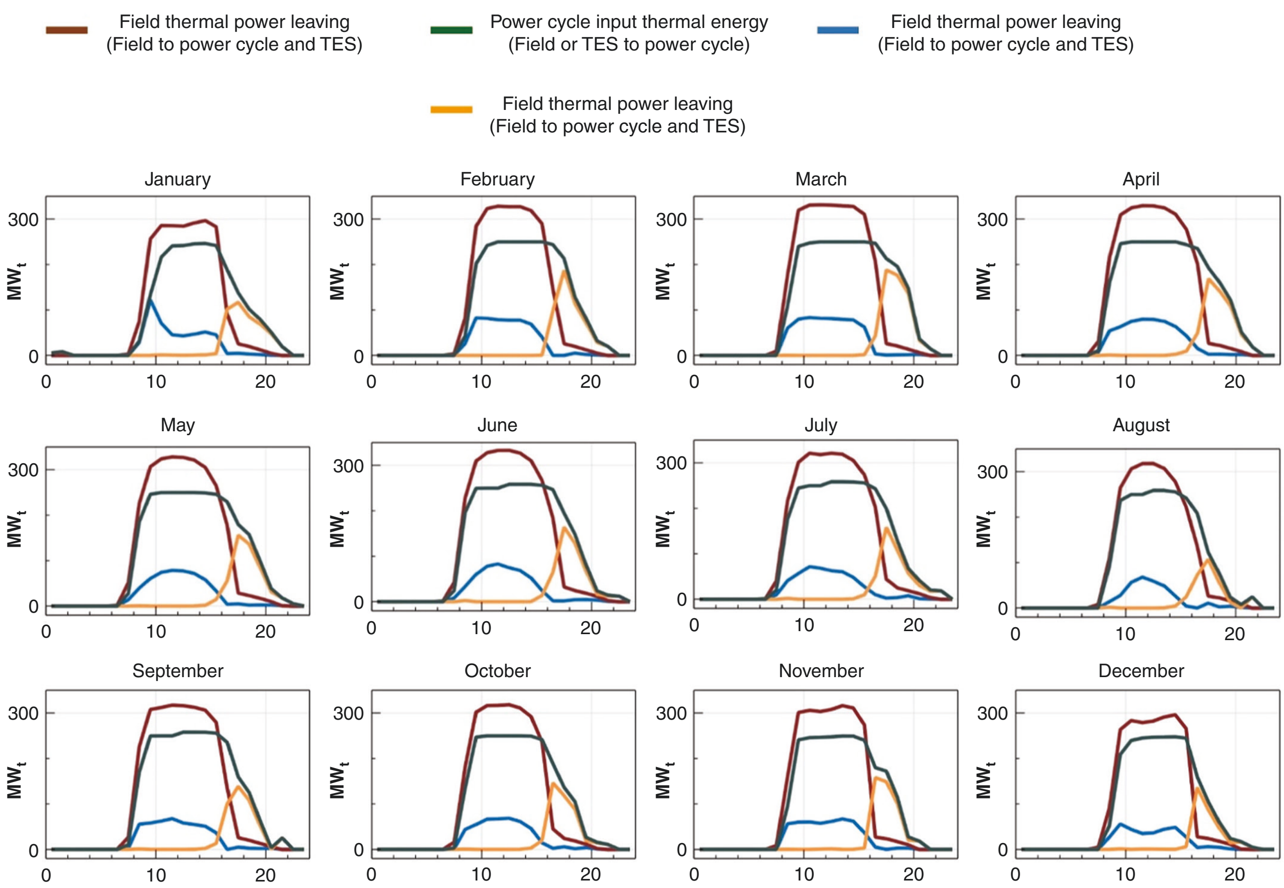

Fig. 15: The flow of thermal energy between different components of the plant for each month of the year.

Table 7: Annual thermal and parasitic losses

Solar field	
Solar energy loss due to tracking errors	1.66×10^5 MWh_t
Electricity required for tracking system	458.15 MWh_e
Electricity required for pumping the HTF	1.48×10^4 MWh_e
Field piping thermal losses	1.17×10^4 MWh_t
Receiver thermal losses	2.29×10^5 MWh_t
TES system	
TES thermal losses	8.386×10^3 MWh_t
TES freeze protection	140.882 MWh_e
Power cycle	
Parasitic energy of TES and cycle HTF pump	2.833×10^3 MWh_e
Parasitic energy of condenser operation	5.689×10^3 MWh_e

power cycle running. In August, this value reaches its minimum limit, which is 9.371×10^3 MW_t. Due to the thermal losses in the tanks, the thermal energy discharged from the TES system is less than the thermal energy stored in it by 4.25% annually.

In addition to the tracking errors, the thermal and parasitic losses (Table 7) are also responsible for reducing the plant's output. Thermal losses occur in the solar field and TES system, whereas parasitic losses represent the electricity generated to cover the plant's requirements. Tracking errors cause a loss of ~8.87% of the solar energy received by the solar field. This loss is unavoidable since the north–south orientation with an east–west tracking axis is considered to have higher efficiency than other tracking techniques. In this case, 2.29×10^5 MWh of thermal energy is lost annually due to receiver thermal losses in the solar field. This represents ~13.45% of the annual solar thermal energy absorbed by the solar field. However, these inevitable thermal losses can be reduced by developing receivers with better performance. Another significant kind of thermal loss occurs in storage tanks. Due to the high operating temperatures of molten salts, 8.386×10^3 MWh of thermal energy is lost annually. In addition, 140.882 MWh of electricity is required annually to avoid freezing in storage tanks. Table 7 summarizes the plant's annual losses and parasitic energy requirements.

3.2 Cost analysis

The results of economic calculations show that the annual capital costs, fixed operating costs and variable operating costs of the proposed plant are \$450 216 256, \$5 280 660 and \$1 125 291, respectively. Also, they show that the LCOE is 0.155 \$/kWh. This value of LCOE is acceptable and proves the economic feasibility of the plant because PTCSP plants with an installed capacity of <100 MW and 4–8 hours of heat storage have a value of LCOE of ~0.2 \$/kWh recorded in 2019 [44]. A more detailed analysis (Fig. 16) demonstrates the dependence of the LCOE on both the solar multiple and the storage duration. Because the solar multiple expresses the solar field size, plants with higher solar multiples can generate more electricity, leading to lower LCOE values. Similarly, plants with longer storage durations have low LCOE due to their high annual output. As shown in Fig. 16, a similar plant with a solar multiple of 5 and a storage duration of 14 hours has an LCOE of ~0.073 \$/kWh, which is less than the LCOE of the proposed plant by 53%.

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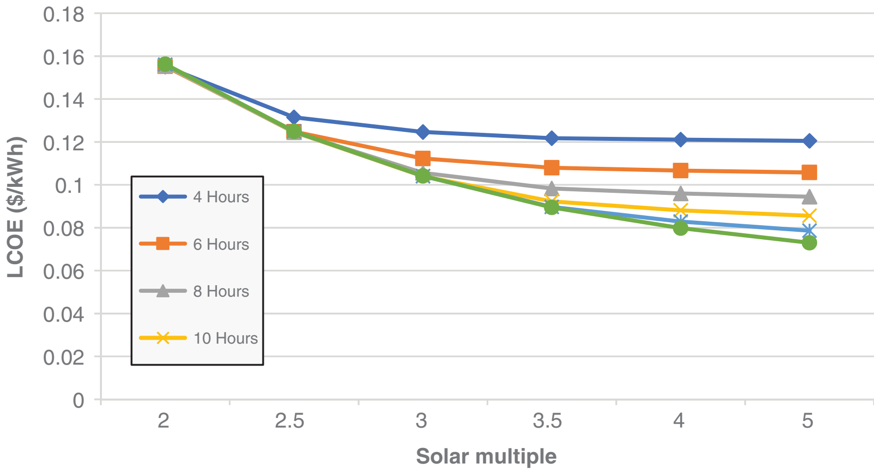

Fig. 16: LCOE as a function of solar multiple and storage duration.

Table 8: Comparison between the proposed plant and some existing PTCSP plants

Name	Location	Nominal capacity (MW_e)	Storage capacity (hours)	Expected annual electricity production (GWh/yr)	Design capacity factor	LCOE (US$/kWh)	Ref.
Aste 1A	Alcázar de San Juan, Spain	50	8	170	38.81%	0.17	[5]
La Florida	Badajoz, Spain	50	7.5	175	39.95%	0.24	[5]
NOOR I	Ouarzazate, Morocco	160	3	370	26.4%	0.28	[5]
NOOR II	Ouarzazate, Morocco	200	7	600	34.25%	0.16	[5]
Solana Generating Station	Phoenix, USA	250	6	944	43.1%	0.2	[5]
Xina Solar One	Pofadder, South Africa	50	5.5	199	45.43%	0.2	[5]
This work	Wadi Halfa, Sudan	80	6	281.145	40.1%	0.155	–

These findings are consistent with the available data from existing similar PTCSP plants. A comparison between the proposed plant and six PTCSP plants located in four different countries (Table 8) shows that the proposed plant has a higher CF than most of the other plants. In addition, the LCOE of the proposed plant is reasonable and less than that of the other plants.

4 Conclusion

This paper has provided a detailed analysis of both the thermal performance and the economic feasibility of a hypothetical PTCSP plant in Sudan using SAM software. The simulation analysis examined 15 locations representing different regions of Sudan to determine the best location to establish PTCSP plants. Researchers and policymakers can use this selection process to determine the potential of CSP technologies in Sudan. In addition, the different parts of the hypothetical PTCSP plant were studied and the properties of fluids used in the plant were taken into consideration. Moreover, a simple analysis was done to estimate the plant's LCOE, and the effects of the solar field's size and TES system capacity on the LCOE were demonstrated. The important findings of this study are the following:

- Wadi Halfa (21.8° N, 31.35° E) is considered the best location to establish the plant. The city has one of the highest DNI rates in Sudan, which is ~2409 kWh/m^2 annually. In addition to that, the city is close to reliable water resources (Nile River) and has the advantage of land availability.
- An 80-MW PTCSP plant placed at this location with a solar multiple of 2 and 6 hours of heat storage was found to have a CF of 40.1% and an overall efficiency of 15%. Therefore, the proposed plant is technically feasible since its CF and overall efficiency are within the range of characteristics of such plants. The proposed plant was found to be able to produce ~281.145 GWh of electricity annually.
- The proposed plant was found to reach its maximum efficiency in March since it produces ~26.41 GWh of electricity in this month of the year. On the other hand, the minimum efficiency was recorded in December when the plant was found to produce ~20.37 GWh of electricity.
- Therminol VP-1 synthetic oil was used as an HTF in the solar field. The use of synthetic oils is suggested to avoid high thermal losses and prevent freezing in the solar field. HITEC solar salt was used as a storage medium because of its high heat capacity and low cost. 140.882 MWh of electricity was found to be required to prevent the freezing of the molten salt in storage tanks.
- Tracking errors are inevitable and cause ~8.87% of the annual received solar thermal energy to be lost. Thermal losses also take place in the solar field and cause a reduction of ~13.45% of the absorbed solar thermal energy annually.
- The cost analysis proved the economic feasibility of the proposed plant as it has an LCOE of 0.155 $/kWh, which is lower than similar plants. Moreover, the LCOE could be reduced more by increasing the solar multiple and the storage duration of the plant. It was found that increasing the solar multiple to 5 and the storage duration to 14 hours per day can reduce the plant's LCOE by 53%.

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5 Future work

A more detailed economic analysis would be useful in future work. Further work could also focus on comparing different CSP technologies to determine which technology has the best performance under the climate conditions of Sudan. Analysing a CSP plant sized to fit the exact electricity load in Sudan may be a suggested topic for a future study. Furthermore, future studies could examine dedicated CSP plant applications, such as seawater distillation and industrial process heating.

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Conflict of interest statement

None declared.

References

[1] International Energy Agency (IEA). *Data and Statistics*. 2022. https://www.iea.org/data-and-statistics (12 February 2022, date last accessed).
[2] International Renewable Energy Agency (IRENA). *Statistics Time Series*. 2020. https://www.irena.org/Statistics/View-Data-by-Topic/Capacity-and-Generation/Statistics-Time-Series (12 February 2022, date last accessed).
[3] Yan H, Wang A, Chong D, *et al*. Review on performance analysis of the power block in concentrated solar power plants. *Energy Technology*, 2021, 9:2000621.
[4] Wang Z. *Design of Solar Thermal Power Plants*. London: Academic Press, 2019.
[5] National Renewable Energy Laboratory (NREL). *Concentrating Solar Power Projects (SolarPACES)*. https://solarpaces.nrel.gov/ (13 February 2022, date last accessed).
[6] TRACKING SDG. *The Energy Progress Report. Sudan Report*. 2020. https://trackingsdg7.esmap.org/country/sudan (12 February 2022, date last accessed).
[7] International Renewable Energy Agency (IRENA). *Final Renewable Energy Consumption*. 2020. https://www.irena.org/Statistics/View-Data-by-Topic/Renewable-Energy-Balances/Final-Renewable-Energy-Consumption (12 February 2022, date last accessed).
[8] International Renewable Energy Agency (IRENA). *Solar Energy*. 2020. https://www.irena.org/solar (13 February 2022, date last accessed).
[9] Abdalla SNM, Özcan H. Design and simulation of a 1-GWp solar photovoltaic power station in Sudan. *Clean Energy*, 2021, 5:57–78.
[10] Abdeen O, Mourad M, Salim H. A comparison study of PV (5MW) based on PVsyst program for evaluation productive energy to connect with the grid. Sudan case study. In: *2019 1st International Conference on Sustainable Renewable Energy Systems and Applications (ICSRESA)*, Tebessa, Algeria, 4–5 December 2019, 1–6.
[11] Abuelyamen A, Mohamed MH. Techno-Economic Study of Installing 10 MW PV Power Plant in Sudan. In: *ASME 2016 International Mechanical Engineering Congress and Exposition*, Phoenix, Arizona, USA, 11–17 November 2016.
[12] Bakhit MAE. Comparative study of photovoltaics and dish Stirling engines for decentralised power generation in Sudan. *Master's thesis*. Universiti Kebangsaan Malaysia, Bangi, Malaysia, 2014, 1–87.
[13] Azouzoute A, Alami Merrouni A, Touili S. Overview of the integration of CSP as an alternative energy source in the MENA region. *Energy Strategy Reviews*, 1004, 2020:93.
[14] Mohamed A, Yahiya A. Development and characterization of parabolic trough for generating 50MW concentrating solar power plant. In: *2016 Conference of Basic Sciences and Engineering Studies (SGCAC)*, Khartoum, Sudan, 20–23 February 2016, 80–87.
[15] Ibrahim SHA, Khayal OMES. Solar energy potential and assessment of CSP plant accounting for sustainability in Sudan. *International Journal of Engineering Applied Sciences and Technology*, 2020, 5:13–19.
[16] Asiri IM, Suliman AY. Design and analysis of parabolic trough collector power plant in Saudi Arabia. *International Transaction Journal of Engineering, Management, Applied Sciences & Technologies*, 2021, 12:12A–12F.
[17] Benhadji Serradj DE, Sebitosi AB, Fadlallah SO. Design and performance analysis of a parabolic trough power plant under the climatological conditions of Tamanrasset, Algeria. *Int J Environ Sci Technol*, 2022, 19:3359–3376.
[18] Bishoyi D, Sudhakar K. Modeling and performance simulation of 100MW PTC based solar thermal power plant in Udaipur India. *Case Studies in Thermal Engineering*, 2017, 10:216–226.
[19] Liaqat K, Anss M, Ali A, *et al*. Modeling and simulation of a 100 MW concentrated solar thermal power plant using parabolic trough collectors in Pakistan. In: *1st International Conference on Advances in Engineering and Technology (ICAET-2018)*, IOP Conference Series: Materials Science and Engineering, Quetta, Pakistan, 2–3 April 2018, 414:012032.
[20] De Oliveira Siqueira AM, Altabash GA, Barhouche RF, *et al*. Simulation study of parabolic trough solar power plants in Brazil. *International Journal of Research: GRANTHAALAYAH*, 2019, 7:17–28.
[21] Kordmahaleh AA, Naghashzadegan M, Javaherdeh K, *et al*. Design of a 25 MWe solar thermal power plant in Iran with using parabolic trough collectors and a two-tank molten salt storage system. *Int J Photoenergy*, 2017, 2017:14210184–14210111.
[22] National Renewable Energy Laboratory. *System Advisor Model Version 2020.11.29*. https://sam.nrel.gov/download/version-2021-12-01.html (12 March 2022, date last accessed).
[23] Ezeanya EK, Massiha GH, Simon WE, *et al*. System advisor model (SAM) simulation modelling of a concentrating solar thermal power plant with comparison to actual performance data. *Cogent Engineering*, 2018, 5:1524051.
[24] National Renewable Energy Laboratory. *System Advisor Model (SAM): CSP validation*. https://sam.nrel.gov/concentrating-solar-power/csp-validation.html (5 May 2022, date last accessed).
[25] Boretti A, Nayfeh J, Al-Kouz W. Validation of SAM modeling of concentrated solar power plants. *Energies*, 2020, 13:1949. https://doi.org/10.3390/en13081949 (5 May 2022, date last accessed).
[26] Mills S. Combining solar power with coal-fired power plants, or cofiring natural gas. *Clean Energy*, 2018, 2:1–9.
[27] Schlecht M, Meyer R. 4: Site selection and feasibility analysis for concentrating solar power (CSP) systems. In: Lovegrove K, Stein W (eds). *Concentrating Solar Power Technology*. Cambridge: Woodhead Publishing, 2012, 91–119.
[28] Global Solar Atlas. *Map and Data Downloads (Sudan)*. https://globalsolaratlas.info/download/sudan (21 February 2022, date last accessed).
[29] Blanco MJ, Miller S. 1: Introduction to concentrating solar thermal (CST) technologies. In: Blanco MJ, Santigosa LR (eds).

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Advances in Concentrating Solar Thermal Research and Technology. Cambridge: Woodhead Publishing, 2017, 3–25.
[30] Fernández-García A, Zarza E, Valenzuela L, *et al*. Parabolic-trough solar collectors and their applications. *Renew Sustain Energy Rev*, 2010, 14:1695–1721.
[31] Moya EZ. 7: Parabolic-trough concentrating solar power (CSP) systems. In: Lovegrove K, Stein W (eds). *Concentrating Solar Power Technology*. Cambridge: Woodhead Publishing, 2012, 197–239.
[32] Kalogirou SA. *Solar Energy Engineering: Processes and Systems*. 2nd edn. London: Academic Press, 2013.
[33] Abed N, Afgan I, Cioncolini A, *et al*. Assessment and evaluation of the thermal performance of various working fluids in parabolic trough collectors of solar thermal power plants under non-uniform heat flux distribution conditions. *Energies*, 2020, 13:3776.
[34] Fredriksson J, Eickhoff M, Giese L, *et al*. A comparison and evaluation of innovative parabolic trough collector concepts for large-scale application. *Sol Energy*, 2021, 215:266–310.
[35] SCHOTT Solar. *SCHOTT PTR®70 Receiver The 4th Generation, Brochure*. 2013: Germany. https://d3pcsg2wjq9izr.cloudfront.net/files/31497/download/461508/1-schott_ptr70_4th_generation_brochure.pdf (5 March 2022, date last accessed).
[36] Wagner MJ, Gilman P. *Technical Manual for the SAM Physical Trough Model (No. NREL/TP-5500-51825)*. Golden, CO: National Renewable Energy Laboratory (NREL), 2011. https://www.nrel.gov/docs/fy11osti/51825.pdf (20 February 2022, date last accessed).
[37] Lovegrove K, Pye J. 2: Fundamental principles of concentrating solar power (CSP) systems. In: Lovegrove K, Stein W (eds). *Concentrating Solar Power Technology*. Cambridge: Woodhead Publishing, 2012, 16–67.
[38] Lopes T, Fasquelle T, Silva HG. Pressure drops, heat transfer coefficient, costs and power block design for direct storage parabolic trough power plants running molten salts. *Renew Energy*, 2021, 163:530–543.
[39] Therminol Heat Transfer Fluids by Eastman. *THERMINOL® VP-1 Heat Transfer Fluid*. https://www.eastman.com/Literature_Center/T/TF9141.pdf#_ga=2.257485133.796672539.1650639564-1570428542.1650639564 (16 February 2022, date last accessed).
[40] Caraballo A, Galán-Casado S, Caballero A, *et al*. Molten salts for sensible thermal energy storage: a review and an energy performance analysis. *Energies*, 2021, 14:1197.
[41] International Renewable Energy Agency (IRENA). *Renewable Energy Technologies: Cost Analysis Series*. Abu Dhabi: International Renewable Energy Agency, 2012.
[42] National Renewable Energy Laboratory. *System Advisor Model Version 2020.11.29 (SAM 2020.11.29) User Documentation. LCOE Calculator*. https://samrepo.nrelcloud.org/help/index.html?fin_lcoefcr.htm (20 February 2022, date last accessed).
[43] Boretti A, Castelletto S. Techno-economic performances of future concentrating solar power plants in Australia. *Humanities and Social Sciences Communications*, 2021, 8:326.
[44] International Renewable Energy Agency (IRENA). *Renewable Power Generation Costs in 2020*. Abu Dhabi: International Renewable Energy Agency, 2021.

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